DE–7220 **11**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is ICT?
2. What is Groupware?
3. What is MODEM? What is its functions?
4. What is meant by Video Conferencing?
5. What are the bitwise operations in C?
6. What are the logical operators in C?
7. What is pointer? Mention its advantages.
8. How do you store the two-dimensional array in the memory?
9. What is recursion?
10. Define Macro. Give examples.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. What is an application software? Explain the various types of application software.
12. Explain briefly about internet web browsers.
13. Explain briefly about local area networks.

14. Explain the switch-case constructions with an example.

15. Explain the concept of structure with an example program.

16. Explain the concept of modularisation.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions

17. Describe in detail the various milestones in Information Technology.

18. Explain with examples, various structures in C.

19. Explain the concepts of pass-by-value and pass-by-reference with examples.

DE–7221 **12**

DISTANCE EDUCATION

M.Sc. (I.T) DEGREE EXAMINATION, DECEMBER 2013

OPERATING SYSTEMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Draw the abstract view of the components of a computer system.

2. Identify the memory management functions that exhibited by an operating system.

3. Distinguish CPU burst cycle from I/O burst cycle.

4. Define :

   (a) Counting semaphore

   (b) Binary Semaphore.

5. What is an assignment edge? How it differs from request edge? Illustrate.

6. What is meant by double buffering?

7. Differentiate external fragmentation from internal fragmentation.

8. What is an elevator algorithm? Why it is called so?

9. What is meant by pure demand paging?

10. Define : “TLB miss”.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss the concept of operating system structure.

12. Write about Semaphores.

13. Narrate the priority and round robin scheduling techniques adopted in a CPU.

14. Explain the deadlock prevention methodology.

15. Write a note on contiguous memory management concept.

16. Elaborate on efficiency and performance of file system implementation.

PART C – (2 × 15 = 30 marks)

Answer any TWO questions.

17. Give a neat description on operating system calls.

18. With an example, explain the importance of inter process communication.

19. Explain the concept of paging with fixed and variable memory partitioning.

DE–7222 **13**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013

OBJECT ORIENTED PROGRAMMING AND C++

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define an Object. Give an example.
2. What is polymorphism?
3. What are the user-defined data types available in C++?
4. What is a scope resolution operator?
5. What are overloaded constructors?
6. Explain the use of destructors.
7. What is a function call operator?
8. Explain class member access operator.
9. What is multilevel inheritance?
10. What is the use of a virtual function?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the benefits of object oriented programming.
12. Explain the two method for creating symbolic constants in C++.

13. Explain the concept of function overloading with an example.

14. Explain the use of copy constructor with an example.

15. Explain overloading of binary operators in C++.

16. Explain hybrid inheritance with an example.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Describe the basic concepts of object oriented programming.

18. Explain the various control structures in C++ by giving examples.

19. Explain the operators available for abstract data type conversion.

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DE–7223 **14**

DISTANCE EDUCATION

M.Sc.(IT) DEGREE EXAMINATION, DECEMBER 2013.

DATA STRUCTURES AND ALGORITHMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What are the properties of an algorithm?

2. What is data structure?

3. What is the difference between an array and a linked list?

4. List any three applications of 'Stack'.

5. What are the operations that can be performed on a string?

6. Define : string and an empty string.

7. If a tree consists of '3 nodes' then how many different tree representations are possible?

8. What is DAG?

9. What is meant by heap data structure?

10. Distinguish between linear and binary search.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write a note on complexity measures of algorithms.

12. What is stack? Explain the operations on stack.

13. Write about pattern matching algorithms.

14. Explain the representation and traversal of graph.

15. Write the bubble sort algorithm and explain it with an example.

16. Write a note on 'Tree searching'.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Compare and contrast singly linked lists with doubly linked lists.

18. Explain quick sort with an example.

19. Discuss the need and types of hashing.

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DE–7224 **15**

DISTANCE EDUCATION

M.Sc.(IT) DEGREE EXAMINATION, DECEMBER 2013.

OBJECT ORIENTED DBMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is database schema?
2. State the levels of abstraction in DBMS.
3. List the symbols used in ER model.
4. What is QBE?
5. What is the difference between Boyce–codd and Third normal form?
6. What is meant by 'database tuning'?
7. List the access control methods in DBMS security.
8. What is DTD?
9. Write the use of XML.
10. Write the ODMG Data model.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the different data models.
12. Describe the operations of relational algebra.

13. Explain the concept of 'indexing' in physical database design.

14. Compare parallel and distributed databases.

15. Explain the various set operations in SQL.

16. What is WWW?

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. (a) Explain the structure of SQL with examples.

(b) Write a note on 'Triggers'. (8+7)

18. Compare and contrast OODBMS and ORDBMS.

19. Elaborate on the importance of normalization in database design.

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DE–7225 **21**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

INTERNET PROGRAMMING AND WEB DESIGN

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What is meant by Intranet?

2. Define : WWW.

3. List the feature of Java.

4. What is Perl?

5. Write a note on : For loop in Javascript.

6. How to embed VBScript code in an HTML Document? Give an example.

7. What is SGML?

8. What is Microsoft Internet Explorer?

9. What is DTD?

10. What is dynamic graphics?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write about security and encryption.

12. What is CGI? Explain.

13. Write about Microsoft Internet Implementation.

14. Write a VBScript to find the Bigger number among TWO input numbers.

15. How to create tables using HTML? Explain with an example.

16. Write short notes on : Site and Page Design.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain WWW Design issues.

18. Write a JavaScript to find the sum and average of N input numbers.

19. Explain creating a custom integrated application with multiple protocols.

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DE–7226 **22**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER NETWORKS

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

All questions carry equal marks.

1. What do you mean by unicasting and multicasting?

2. What do you understand by 'Internet' and 'Intranet'?

3. What are the three components of an optical transmission system?

4. What is ment by bit stuffing?

5. Write down the responsibilities of Data Link Layer.

6. How does a path differ frau a route?

7. Name any two interfaces used in ATM networks.

8. List out the duties of Transport Layer.

9. Write down any two differences and similarities of TCP and UDP.

10. What do DNS stand for?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions only.

11. Write a note on transmission modes.

12. Explain the role of unguided media in transmission.

13. Describe the mechanism behind the collision—free protocols.

14. How does the network layer ensure the delivery of packets from the source to destination?

15. Explain the various QoS parameters linked transport layer.

16. How authentication achieved thru Kerberos?

PART C — (2 × 15 = 30 marks)

Answer any TWO questions only.

17. With necessary diagram, explain the OSI network model.

18. With an example database, Explain the concept of link state routing in detail.

19. Write an essay on "Ways to enhance the Network Security".

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DE–7227 **23**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

SOFTWARE ENGINEERING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Mention the activities of linear sequential model.

2. Who proposed MOI model of leadership? What does MOI stand for?

3. Distinguish Full-experience components from partial-experience components.

4. List four types of risk components.

5. Identify four different degrees of rigor.

6. Define Appraisal costs. Give examples.

7. What is meant by control hierarchy?

8. What are the three generic classes of methods and tools to conduct rapid prototyping?

9. Define :

   (a) White box testing

   (b) Black box testing.

10. Define the cyclomatic complexity for a flow graph, G.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give a note on software process.

12. Write about COCOMO model in detail.

13. Discuss the concept of software quality assurance.

14. Narrate the effective modular design that enhances the software design.

15. Summarize the unit testing procedure.

16. Discuss on the importance of Debugging.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Give a detailed description on various evolutionary software process models.

18. (a) Discuss the process of statistical software quality assurance. (8)

    (b) Write about system engineering hierarchy. (7)

19. Narrate the procedure of white box testing with basis path testing strategies.

DE–7228 **24**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, DECEMBER 2013.

Information Technology

VISUAL PROGRAMMING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are DLLs?

2. List out the SDK tools with a brief description of each of them.

3. What is the main difference between procedures and functions in VB?

4. What do you mean by SDI forms?

5. What is the purpose of tool box in VB? List out the important controls in the tool bar.

6. What is a record set?

7. Differentiate classes and objects in VC++.

8. What is MFC? Mention its purpose.

9. What is an exception? Give an example.

10. What is OLE? What is the use of it?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the data types in windows programming.

12. Write a note on context help.

13. Explain tile important properties of forms in VB.

14. Explain how check boxes and option buttons may be used with suitable examples.

15. What is serialization in VC++? Explain with an example.

16. Explain how exceptions are handled in VC++ with suitable examples.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the following :

    (a) Functions in VB

    (b) Dialog boxes in VB.

18. Discuss event handling in VC++. Give suitable examples.

19. Explain how a database can be accessed in a VC++ application, with an appropriate example.

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DE–7229 **25**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

MULTIMEDIA AND APPLICATIONS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is meant by multimedia? What are its components?
2. Expand the following acronyms : (a) XA and (b) SFBI.
3. What do you understand by bi tone images?
4. What is MIDI? What it does?
5. What is data compression?
6. List out the parameters to be considered while evaluating a compression system.
7. Define Aspect Ratio.
8. What is an object with respect to MM? Give examples.
9. Mention the role of MIME in networks.
10. Write down any three popular TV standards.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions only.

11. Explain the key areas of applications of multimedia.
12. Write a note on commercial tools used in multimedia development.

13. Explain the various colour models.

14. How does DVI technology differ from other standards?

15. Present a note on Multimedia on networks.

16. Write a note on the role of object Orientation in multimedia development.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions only.

17. Explain the various components of multimedia system development.

18. Elaborately discuss the standards behind in image compression.

19. Write an essay on the importance of multimedia in medicine.

DE–7633 **16**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

LAB I — C++ LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Break up of marks :

Record : 10
Algorithm: 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

1. Write a C++ program using friend function which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

---------- Cut here ----------

2. Write a C++ program which will accept a string of 10 characters in length from the keyboard and count the occurrences of each of the five vowels in the string. The output should be in a (tapped) format as given in this example.

A	E	I	O	U
0	1	0	0	1

3. Write a program in C++ to reverse a string and to find whether it is a palindrome.

---------- Cut here ----------

4. Write a C++ program to process students marks with the help of classes. The class has private variables for name, mark1, mark2, mark3. It has 2 member functions getdata ( ) to get input and result ( ) to print the results. All subject marks must be > = 50 for PASS otherwise fail.

---------- Cut here ----------

5. Create a class DONOR that contains donor number, donor name, age, address, sex, blood group.

   Write a menu driven C++ program to display the number, name and address of the donors for the following categories

   (a) Blood donors with blood group O+

   (b) Blood donors in the age group between 16 and 25

   (c) Female donors having blood group A in the age between 19 and 24.

---------- Cut here ----------

6. Create a class called employee that contains Employee number, Employee name, Basic pay, designation, deductions (LIC, PF). Include a member function to getdata from user for 'n' employees. Write a C++ program to prepare payslips for n employees using the following details.

   DA = 40% of basic pay

   HRA = 20% of basic pay

   GPay = Basic pay + DA + HRA

   NPay = Gpay – Deductions.

   Print the O/P in a neat format.

7. Raising a number n to power P is the same as multiplying by itself P times. Write a function called power ( ) that takes a double value for n and an int value for P, and returns the result as double value. Use default argument of 2 for P, so that if this argument is omitted, the number will be squared. Write a main ( ) function that gets values from the user to test this function.

---------- Cut here ----------

8. Create an equivalent of a four-function calculator. The program should request the user to enter a number, an operator and another number. It should then carryout the specified arithmetic operations : adding, subtracting, multiplying, dividing the 2 numbers. Finally, it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

DE–7634 **17**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, DECEMBER 2013.

LAB – II: Data Structures

DATA STRUCTURES LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break – up of marks

Record:	10
Algorithm:	10
Program :	50
Debugging:	10
Execution :	10
Result :	10
Total :	100

ONE question should be selected by the Examiner and given to the candidates by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C++ program to add and subtract the two given mxn matrices.

   (b) Write a C++ program for creation of linked list and performing its operation using array.

---------- Cut here ----------

2. (a) Write a C++ program to add and multiply the two given mxn matrices.

   (b) Write a C++ program for implementing stack operations using array.

3. (a) Write a C++ program to search an element using binary search.

   (b) Write a C++ program to copy the content of one file to another file.

---------- Cut here ----------

4. (a) Write a C++ program to sort the following list using bubble sort.
   -12, 25, -45, 0, 42, -32, -27, -20, 14

   (b) Write a C++ program for implementing queue structure and its operations using array.

---------- Cut here ----------

5. (a) Write a C++ program to sort elements using insertion sort.

   (b) Write a C++ program for creation of linked list and performing its operations using pointers.

DE–7635 **26**

DISTANCE EDUCATION

M.Sc.(Information Technology) DEGREE EXAMINATION, DECEMBER 2013.

INTERNET PROGRAMMING LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break up of Marks

Record Note Book :	10
Algorithm :	10
Program :	50
Debugging :	10
Execution :	10
Result :	10
Total :	100

Examiner has to choose any ONE question.

1. (a) Write a Java Script to create a window by using the confirm message.

   (b) Write a Java Program for inserting and deleting records from a student table using database connectivity.

---------- Cut here ----------

2. (a) Write a VBScript to do the following

   (i) Find the difference between the two dates.

   (ii) Find the age of a person by providing date of birth.

   (b) Write a Java program using Applet to create the frames and its controls.

3. (a) Write a Java Script to create a color Palette and display the background in the color chosen from the palette.

(b) Design a web page using HTML code shows your biodata.

---------- Cut here ----------

4. (a) Write a VB Script to do the following

(i) Check the given password

(ii) Change the existing password.

(b) Write a Java program to display the records in an employee table using data base connectivity.

---------- Cut here ----------

5. (a) Write a Java program using Applet to create the frames and its controls.

(b) Design a Web page using HTML code shows Employee's salary details.

---------- Cut here ----------

6. (a) Write a swing program to create Buttons with

(i) Tool tip text

(ii) Image

(iii) Border

(iv) Short cut key

(b) Design a web page using HTML code shows your biodata.

7. (a) Write a Java script to create an order form to select the home appliances.

(b) Design a web page using HTML code to display the details of your institution where you are doing course.

---------- Cut here ----------

8. (a) Write a Java program using Applet to display 3 images when 3 buttons in the Border Layout are clicked. The image should be displayed at the centre.

(b) Write a VB Script to create a calendar for a given month and year.

---------- Cut here ----------

9. (a) Write a VB script to do the following

(i) Display the current date.

(ii) Find the age of a person by providing date of birth.

(b) Write a Java program using Applet to display the dialogue and menu in applet.

---------- Cut here ----------

10. (a) Write an applet program to draw a filled rectangle and circle. Use graphic controls.

(b) Write a swing program to create the Tabbed panels.

DE–7636 **27**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, DECEMBER 2013.

LAB IV — VISUAL PROGRAMMING

(2002 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

1. (a) Write a VB program to create a simple calculator.

   (b) Write a VC++ program to fill background of the client area with a bitmap image.

---------- Cut here ----------

2. (a) Using grid control in VB, display a multiplication table with 20 rows and columns.

   (b) Write a VC++ program to find the number buttons in the mouse, if attached to the system.

---------- Cut here ----------

3. (a) Write a VB program to display the text messages from a given sequence one by one in a text box, each for a duration of 10 seconds.

   (b) Write a VC++ program to get the status of the shift and toggle keys using MFC.

4. (a) Using textbox, combobox and command buttons, write a VB program to include strings in list at runtime, with provisions for sorting and deleting strings.

   (b) Write a VC++ program to generate a status bar and show the status of caps lock, num lock and scroll lock keys in it.

---------- Cut here ----------

5. (a) In VB, write an event procedure to accept two strings as input, and to check if the first string is a substring of the second one or not.

   (b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

6. (a) Write a VB program to insert, delete and search records in a random access file.

   (b) Write VC++ program to create list box in a window.

---------- Cut here ----------

7. (a) In VB, write event procedure to check if a given string is palindrome or not.

   (b) Write a VC++ program to check the status of mouse in the system, ie. whether it is present, and if so, how many buttons it has.

---------- Cut here ----------

8. (a) Create a table with student details for preparing mark sheets, and make provisions for adding, deleting, and searching the table. Write the program in VB.

   (b) Write a VC++ program to fill background of the client area with a bitmap.

9. (a) Write event procedure for the following :

   (i) To display date and time in a text box.

   (ii) To convert given temperature from Fahrenheit to centigrade and vice versa.

   (b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

10. (a) Create a table with employee details for preparing salary slips, and make provisions for adding, deleting and searching the table. Write the program in VB.

    (b) Write a VC++ program to create a list box in a window.